

## INVENTION DISCLOSURE FORM FOR PATENTS

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### 1. Particulars of Inventors

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### 2. Provide title of the invention: In 100 words or less, please provide an abstract or summary of the invention:

#### ➤ An Intelligent Multimodal System for QR Code Phishing Detection and Mitigation

**Abstract:** This invention presents an intelligent multimodal system for detecting QR code phishing attacks in real time. The system extracts encoded URLs and analyzes QR image characteristics, URL patterns, domain information, redirection behavior, security certificates, and webpage content. A machine learning based fusion engine combines these features to generate a dynamic Quishing Risk Score. Based on the calculated risk, the system classifies each QR code as legitimate, suspicious, or malicious. It provides explainable warnings and automatically permits, blocks, or isolates access before redirection. The proposed approach improves early detection, reduces false alerts, and protects users from credential theft and financial fraud.

### 3. Detail description of the invention:( Answer to all below are required in detail)

#### a. Problem the invention is solving:

- QR codes hide the destination URL, preventing users from checking it before scanning.
- Attackers manipulate or replace genuine QR codes with malicious QR codes.
- Malicious QR codes redirect users to fake login, banking or payment websites.

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- Shortened URLs and multiple redirections make malicious destinations difficult to identify.
- Existing tools mainly analyse URLs after users open them.
- Current systems may not jointly analyse the QR image, URL, domain and webpage.
- Newly created phishing domains may bypass signature-based security systems.
- Users may suffer credential theft, financial fraud, malware infection and data loss.
- The invention analyses threats before allowing access to the destination.
- It warns users and blocks or isolates suspicious and malicious QR-code links.

**b. General Utility/application of the invention:**

- **Banking and Digital Payment Security:** The invention may be integrated with banking, UPI and digital-wallet applications to analyse payment QR codes before completing transactions. It can identify manipulated QR codes, suspicious payment links, unauthorised redirections and fake merchant destinations, thereby protecting users from financial fraud.
- **Mobile QR Scanner and Camera Applications:** The invention can be incorporated into smartphone cameras and QR-scanning applications. Before opening an encoded link, the system analyses the QR image, embedded URL, domain details, security certificate and redirection behaviour. It then allows, warns or blocks access based on the generated risk score.
- **Enterprise and Organisational Cybersecurity:** Organizations may deploy the invention to examine QR codes present in emails, documents, posters, employee identity cards and internal communication systems. It can prevent employees from accessing fake login portals, credential-stealing websites and malicious downloads through fraudulent QR codes.
- **Retail, E-Commerce and Public Services:** The invention can protect QR codes used in restaurants, retail stores, advertisements, ticketing systems, product packaging and public-service platforms. It verifies whether the QR code leads to an authorised destination and warns users when physical replacement, visual manipulation or suspicious redirection is detected.
- **Email, Browser and Web Security:** The invention may be integrated with email gateways, web browsers and browser extensions to detect QR codes embedded in emails and webpages. It can extract and analyse their destinations before users scan or access them, reducing QR-based phishing attacks delivered through digital communication.
- **Authentication and Access-Control Systems:** The invention can secure QR codes used for login, device pairing, account activation, attendance and identity verification. It detects unauthorised or cloned QR codes and prevents users from submitting credentials or authentication information to fraudulent systems.
- **Cybersecurity Monitoring and Incident Response:** Security teams may use the invention to monitor phishing campaigns and analyse suspicious QR-code samples. The generated risk score, detection reasons, URL-redirection details and domain indicators can assist investigators in identifying attack patterns, blocking malicious infrastructure and responding to incidents.

**c. Advantages of the invention disclose the increased efficiency/efficacy :**

- **Early Detection Before Redirection:** The invention analyses the QR code and its embedded destination before the webpage is opened. This prevents users from reaching phishing websites and reduces the possibility of credential theft, financial fraud and malware infection.
- **Multimodal Threat Analysis:** The invention jointly examines QR-image characteristics, URL patterns, domain information, SSL/TLS certificates, redirection behaviour and webpage content. Combining multiple data sources improves detection effectiveness compared with systems that analyse only the encoded URL.

- **Improved Detection Accuracy:** The multimodal fusion engine combines different security indicators to identify both known and previously unseen quishing attacks. This increases detection accuracy and reduces dependence on static blacklists and predefined phishing signatures.
- **Reduced False Alerts:** The invention generates a dynamic Quishing Risk Score by evaluating the combined importance of multiple indicators. This avoids classifying a QR code as malicious based on only one weak indicator and helps reduce false-positive and false-negative results.
- **Detection of Shortened and Redirected URLs:** The system expands shortened URLs and traces multiple redirection paths to determine the final destination. This improves its ability to detect attackers who conceal phishing websites behind URL-shortening services or intermediate webpages.
- **Detection of QR-Code Manipulation:** The image-analysis component can identify visual tampering, overlays, replacement patterns and other suspicious QR-code characteristics. This provides additional protection against physically or digitally manipulated QR codes.
- **Real-Time Automated Protection:** Based on the calculated risk level, the invention automatically allows, warns, blocks or isolates the QR-code destination. This reduces manual security analysis and enables an immediate response to suspicious QR codes.
- **Explainable Detection Results:** The invention provides reasons for its classification, such as a newly registered domain, suspicious URL structure, certificate issue, excessive redirections or webpage imitation. This improves user awareness and assists security personnel in verifying detected threats.
- **Protection Against Emerging Attacks:** By using behavioural, contextual and content-based indicators, the invention can detect newly created phishing websites that may not be available in existing blacklist databases. This improves protection against zero-day and rapidly changing quishing campaigns.
- **Scalable and Flexible Deployment:** The invention can be deployed as a mobile application, browser extension, security API, cloud service or enterprise security solution. It can analyse multiple QR-code samples automatically, thereby increasing operational efficiency across different environments.

**d. Best way of using the invention as well as possible variants :**

**Best Way of Using the Invention:**

- The invention is preferably implemented as a mobile QR-code scanning application.
- The user scans a QR code using the smartphone camera.
- The system decodes the QR code without immediately opening the embedded URL.
- It analyses the QR-code image for possible tampering, replacement, overlays and structural abnormalities.
- The extracted URL is examined for suspicious keywords, shortened links, misleading domains, unusual characters and IP-address-based links.
- The system traces redirection chains to determine the actual final destination.
- Domain age, DNS information, hosting details, reputation and SSL/TLS certificate status are verified.
- The destination webpage is examined in a secure environment for fake login forms, payment requests, malicious scripts and brand impersonation.
- The multimodal fusion engine combines the QR-image, URL, domain, redirection and webpage features.
- A dynamic Quishing Risk Score is generated to classify the QR code as legitimate, suspicious or malicious.

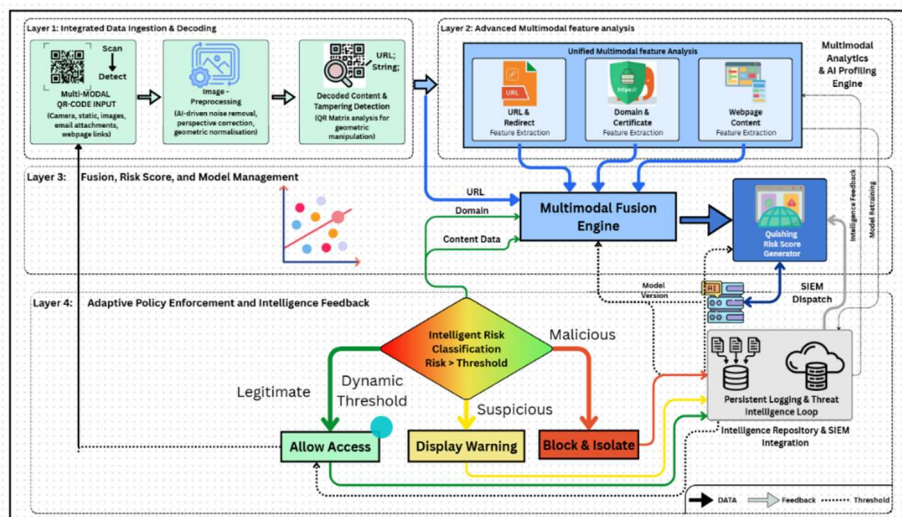
- Based on the risk level, the system automatically allows the link, displays a warning or blocks and isolates the destination.
- The user receives an explainable alert showing the reasons for classifying the QR code as suspicious or malicious.

#### Possible Variants of the Invention:

- **Mobile Application Variant:** Integrated into a smartphone camera or dedicated QR-scanning application.
- **Browser Extension Variant:** Detects and analyses QR codes displayed on websites, emails and online documents.
- **Banking Application Variant:** Verifies payment QR codes, merchant identities and payment destinations before authorising transactions.
- **Digital-Wallet Variant:** Protects UPI and wallet users against fraudulent payment requests and replaced merchant QR codes.
- **Enterprise Security Variant:** Scans QR codes contained in organisational emails, attachments, documents and internal portals.
- **Cloud-Based Variant:** Provides scalable QR-code analysis through a centralized cloud platform or security API.
- **Offline Variant:** Performs lightweight QR-image and URL analysis directly on the device without requiring internet access.
- **Hybrid Variant:** Performs quick analysis locally and sends complex domain and webpage checks to a secure cloud service.
- **Email-Gateway Variant:** Automatically extracts and analyses QR codes contained in incoming emails and attachments.
- **Web-Service/API Variant:** Allows third-party applications to submit QR-code images or URLs for security verification.
- **Kiosk and Point-of-Sale Variant:** Verifies QR codes used at retail counters, ticketing machines and public-service kiosks.
- **Authentication Variant:** Protects QR codes used for account login, device pairing, identity verification and access control.

#### e. Working of invention along with Drawing, schematics and flow diagrams if required with complete explanations:

##### System Architecture:



### Step-by-Step Working Explanation:

#### 1. Technical Field & Innovation Summary

The disclosed system provides a multi-layered, artificial intelligence-driven mechanism for real-time identification and mitigation of QR code-based phishing (Quishing) threats. By integrating physical image tampering detection with deep multimodal feature evaluation, the system evaluates security risks across isolated data streams (visual, structural, network, and behavioural) and executes automated containment policies before end-user exposure occurs.

#### 2. Multi-Layer Architectural Breakdown

The architecture is structured into four primary functional layers (Layers 1–4), operating sequentially and concurrently to form a closed-loop intelligence architecture:

##### Layer 1: Integrated Data Ingestion & Physical Preprocessing

- **Multimodal QR-Code Input Module:** Serves as the primary entry interface, ingesting optical payloads from dynamic sources (live camera sensors) and static digital files (email attachments, uploaded images, rendering DOM elements on web pages).
- **Image Preprocessing Module:** Normalises raw optical data using computer vision algorithms (e.g., contrast restoration, noise suppression, perspective warping, and geometric alignment) to ensure high decoding fidelity under suboptimal capture conditions.
- **Decoded Content & Tampering Detection Module:** Extracts the embedded string payload (URL/URI) while simultaneously analysing the physical QR matrix for physical anomalies (e.g., physical sticker overlays, altered finder/alignment patterns, pixel manipulation).

##### Layer 2: Advanced Multimodal Feature Analysis Engine

- **Unified Multimodal Feature Extraction Pipeline:**
  1. **URL & Redirect Analysis:** Performs static and dynamic path extraction, unrolls shortened links, resolves obfuscated URIs, and traces multi-hop HTTP/HTTPS redirection chains.
  2. **Domain & Certificate Analysis:** Inspects domain registration age, WHOIS attributes, SSL/TLS certificate validity, certificate authority trust chains, and cross-checks active threat databases/DNS blocklists.
  3. **Webpage Content Analysis:** Renders target destinations in a sandboxed, headless environment to extract visual features (e.g., logo cloning detection via OCR/computer vision) and textual features (e.g., natural language processing for credential-harvesting prompts).

##### Layer 3: Multimodal Fusion & Risk Score Engine

- **Multimodal Fusion Engine:** Integrates disparate risk signals (visual, network, structural, and behavioral metrics) into a unified feature vector matrix, resolving conflicting or ambiguous risk signals across individual channels.
- **Quishing Risk Score Generator:** Passes the unified feature matrix through a trained machine learning classification model to output a continuous, normalized risk score parameter ( $R \in [0, 1]$ ).

##### Layer 4: Adaptive Policy Enforcement & Threat Intelligence Feedback

- **Intelligent Risk Classification Router:** Compares the generated risk score against dynamic security thresholds to determine enforcement routing:
- **Allow Access (Legitimate):** Routed when  $R < Threshold_{low}$  granting the user seamless navigation to the destination.
- **Display Interstitial Warning (Suspicious):** Routed when  $Threshold_{low} \leq R < Threshold_{high}$  presenting an inline warning detailing flagged risk indicators.

- Block and Isolate (Malicious): Routed when  $R \geq \text{Threshold}_{\text{high}}$ , actively dropping the session, isolating the network path, and neutralising execution.
- Persistent Logging & Threat Intelligence Loop: Archives all operational metadata, session indicators of compromise (IOCs), and classification outcomes. This module actively feeds telemetry back into the system to retrain AI/ML models and dispatches real-time updates to enterprise SIEM/SOAR platforms.

**4. Have you conducted Primary Patent Search? Yes / No (if yes, attach the patent search results)**

➤ We already checked in IPR Portal, USPTO and Google Patent

**5. List out the known ways about how others have tried to solve the same or similar problems? Indicate the disadvantages of these approaches. In addition, please identify any prior art documentation or other material that explains or provides examples of such prior art efforts**

| S. No. | Existing State of the Art                                                                                                                                                                                                                                                                         | Drawbacks in Existing State of the Art                                                                                                                                                                                                | How the Present Invention Overcomes the Drawback                                                                                                                                     |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | <b>Threshold-based matrix-barcode phishing detection:</b> The system analyses encoded addresses and generates a phishing notification when a calculated value exceeds a threshold. It may warn the user or block website access. Prior art: US9344450B2 – Detecting Phishing of a Matrix Barcode. | The approach is primarily based on encoded address information and threshold-based evaluation. It does not describe comprehensive fusion of QR-image tampering, URL, redirect, domain, certificate and webpage-content features.      | The present invention performs multimodal analysis and combines QR image, URL, domain, redirect, and webpage features to generate a dynamic Quishing Risk Score.                     |
| 2      | <b>Secure data transfer using legitimate QR codes:</b> A legitimate QR code is used for transferring data, and a warning is generated when the transfer is identified as malicious. Prior art: US8296477B1 – Secure Data Transfer Using Legitimate QR Codes.                                      | The approach concentrates on securing data transfer through generated legitimate QR codes. It may not adequately detect externally created, physically replaced or digitally manipulated QR codes encountered in public environments. | The present invention accepts QR codes from physical and digital sources and examines them for overlays, replacement marks, structural irregularities and malicious destinations.    |
| 3      | <b>Cryptographically secured QR-code service:</b> Static or dynamic secure QR codes are generated and verified through a controlled security service to prevent phishing and QR-code manipulation. Prior art: WO2019099089A1 – Secure QR Code Service.                                            | The method generally depends on QR codes being generated and managed within the secured service. Ordinary third-party or legacy QR codes may not contain the required security information.                                           | The present invention analyses both secured and ordinary QR codes without requiring the original issuer to generate a special QR-code format or cryptographic token.                 |
| 4      | <b>Secure QR-code management and replacement protection:</b> A managed QR-code platform controls QR-code destinations and addresses the problem of malicious replacement or redirection. Prior art: US12200151B2 – Systems and Methods for Providing Security for QR Codes.                       | The approach is mainly suitable for QR codes managed within a dedicated platform. Its protection may depend on prior registration, controlled generation or management of the QR code.                                                | The present invention provides scan-time analysis of unknown QR codes using image, URL, domain, redirect and webpage indicators, even when the QR code is not previously registered. |
| 5      | <b>Inbound barcode detection and URL rewriting:</b> QR codes contained in organizational emails or attachments are detected, removed or rewritten so that their URLs can be checked by a scanning service. Prior art: US12542814B2 – Detection of Barcode URL in an Organization Inbound Message. | This approach is primarily intended for inbound organizational communications. It may not protect QR codes scanned from posters, parking meters, restaurant menus, payment counters or other physical locations.                      | The present invention can operate through mobile cameras, scanner applications, browsers, email gateways and enterprise systems, covering both physical and digital QR-code sources. |

|    |                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                           |                                                                                                                                                                                    |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6  | <b>Protected QR scanner using operating-system control:</b> A protected scanner restricts or overrides the normal QR-scanning operation to prevent malicious links from compromising the device. Prior art: US12008105B2 – Protected QR Code Scanner Using Operational System Override.  | The protection is centred on controlling the scanning and destination-opening operation. It does not necessarily provide detailed multimodal threat classification or explainable detection results.      | The present invention analyses multiple threat indicators, assigns a risk category and explains the reasons for warning or blocking the destination.                               |
| 7  | <b>Direct QR-image and pixel-pattern classification:</b> Machine-learning models analyse the structure and pixel patterns of a QR code without extracting its encoded URL. Prior art: Detecting Quishing Attacks with Machine Learning Techniques Through QR Code Analysis.              | Image-only analysis may detect structural patterns but cannot completely evaluate the final URL, domain reputation, redirect chain, certificate condition or phishing webpage content.                    | The present invention combines QR-image analysis with encoded-content and destination analysis, enabling both visual and destination-based threat detection.                       |
| 8  | <b>Lightweight deep-learning classification:</b> A deep-learning model classifies QR-code samples as normal, phishing or malware using QR-image features. Prior art: Detection of QR Code-Based Cyberattacks Using a Lightweight Deep Learning Model.                                    | The reported system has lower recall for the malware category and primarily concentrates on image-based classification. It does not fully analyse live domains, URL redirects and webpage behaviour.      | The present invention combines image classification with live URL, domain, certificate, redirection and webpage analysis to provide broader protection and reduce missed threats.  |
| 9  | <b>QR-image feature extraction using CNN-RNN architecture:</b> Image features are extracted using a convolutional model, reduced through PCA and classified using a recurrent neural network. Prior art: Efficient Malicious QR Code Detection Using an Ensemble Deep Learning Approach. | The approach mainly depends on learned QR-image patterns and training-data coverage. New attacks having visually ordinary QR codes but malicious destination behaviour may remain difficult to identify.  | The present invention examines the actual encoded destination and contextual behaviour in addition to the QR image, improving detection of visually normal but malicious QR codes. |
| 10 | <b>Heuristic and machine-learning URL detection:</b> Encoded URLs are extracted from QR codes and classified using lexical URL, website-structure and JavaScript-related features. Prior art: Detecting Phishing URLs in QR Codes Using Heuristic and Machine Learning Techniques.       | URL-centred detection may fail to detect QR image replacement or tampering. It can also be affected by shortened URLs, multiple redirects, newly registered domains and changes in phishing URL patterns. | The present invention traces redirects, checks the final domain and webpage, analyses QR-image tampering and combines all results through a multimodal fusion engine.              |

**6. List the Technical features and Elements of the invention along with the Description of your invention from start to end.:**

**A. Technical Features and Elements:**

| S. No. | Technical Feature/Element              | Description                                                                                                                                                      |
|--------|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | <b>Multisource QR-Code Acquisition</b> | Accepts QR codes from smartphone cameras, static images, email attachments, webpages and digital documents.                                                      |
| 2      | <b>Controlled QR Decoding</b>          | Decodes the QR content without automatically opening the embedded URL, thereby preventing immediate exposure to a malicious destination.                         |
| 3      | <b>AI-Based Image Preprocessing</b>    | Applies noise removal, perspective correction, rotation correction, resizing, contrast enhancement and geometric normalisation to improve decoding and analysis. |

|    |                                              |                                                                                                                                                                |
|----|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4  | <b>QR Tampering Analysis</b>                 | Examines QR-matrix structure and image characteristics to identify overlays, replacement, distortion and possible geometric manipulation.                      |
| 5  | <b>URL and Redirect Analysis</b>             | Extracts lexical and structural URL features, expands shortened URLs and traces intermediate redirects to identify the final destination.                      |
| 6  | <b>Domain and Certificate Verification</b>   | Checks domain age, DNS information, registration details, domain reputation and SSL/TLS certificate validity.                                                  |
| 7  | <b>Secure Webpage-Content Analysis</b>       | Analyses the destination webpage in an isolated environment for fake login forms, payment requests, brand impersonation, hidden content and malicious scripts. |
| 8  | <b>Multimodal Feature Fusion</b>             | Combines QR-image, decoded-content, URL, redirect, domain, certificate and webpage features into a unified security profile.                                   |
| 9  | <b>Quishing Risk Score</b>                   | Generates a combined numerical score indicating the probability and severity of a QR-code phishing attack.                                                     |
| 10 | <b>Dynamic Risk Threshold</b>                | Adjusts the classification threshold according to the input source, QR-code type, contextual risk, model confidence and threat intelligence.                   |
| 11 | <b>Intelligent Risk Classification</b>       | Classifies each QR code as legitimate, suspicious or malicious based on the risk score and dynamic threshold.                                                  |
| 12 | <b>Adaptive Policy Enforcement</b>           | Automatically allows legitimate destinations, displays warnings for suspicious destinations and blocks or isolates malicious destinations.                     |
| 13 | <b>Explainable Detection</b>                 | Identifies the main reasons for classification, such as suspicious domain, invalid certificate, fake login form or excessive redirection.                      |
| 14 | <b>Persistent Security Logging</b>           | Records the QR-code source, destination, extracted features, risk score, classification and action performed.                                                  |
| 15 | <b>Threat-Intelligence Feedback</b>          | Uses confirmed malicious indicators and earlier detection results to improve future analysis and model performance.                                            |
| 16 | <b>Model Management and SIEM Integration</b> | Supports model versioning, security-model updates and dispatching alerts to an organizational SIEM platform.                                                   |

## B. Description of the Invention from Start to End

1. **QR-Code Collection:** The operation begins when a user scans a QR code or submits a QR-code image obtained from an email, document, webpage or other digital source.
2. **Secure Interception:** The invention intercepts the QR-code content and prevents the embedded destination from opening automatically. The content remains restricted until the security assessment is completed.
3. **Image Enhancement:** The QR image is processed to remove noise and correct blur, rotation, perspective, lighting and geometric distortion.
4. **Decoding and Tampering Detection:** The enhanced QR code is decoded to extract its embedded content. Simultaneously, its image and matrix structure are examined for overlays, replacement, unusual distortion and other manipulation indicators.
5. **Content-Type Identification:** The decoded content is identified as a URL, text string, payment request, authentication link, application-download link or other data type.
6. **URL Feature Analysis:** If a URL is detected, the invention examines its length, special characters, subdomains, suspicious keywords, IP-address usage, encoded characters, brand similarity and URL-shortening services.

7. **Redirection Analysis:** Shortened links are expanded, and intermediate redirects are followed within a controlled environment. The complete redirection chain and final destination are identified.
  8. **Domain Verification:** The final destination is examined using domain age, registration details, DNS information, hosting data, reputation records and similarity with legitimate domains.
  9. **Certificate Verification:** The SSL/TLS certificate is checked for validity, issuer, expiry date and correspondence with the destination domain.
  10. **Webpage Analysis:** The destination webpage is examined in an isolated environment. The invention detects login forms, password fields, payment requests, brand imitation, hidden elements, suspicious scripts and unauthorized downloads.
  11. **Feature Fusion:** The features obtained from the QR image, decoded content, URL, redirect chain, domain, certificate and webpage are combined by the Multimodal Fusion Engine.
  12. **Risk-Score Generation:** The fused feature set is analysed using a machine-learning, deep-learning, ensemble or hybrid model to calculate the Quishing Risk Score.
  13. **Dynamic Threshold Selection:** The system determines an appropriate threshold based on contextual conditions. Higher security sensitivity may be applied to QR codes associated with banking, payments, authentication and unknown sources.
  14. **Risk Classification:** The generated score is compared with the selected threshold:
    - A low-risk QR code is classified as **legitimate**.
    - A medium-risk QR code is classified as **suspicious**.
    - A high-risk QR code is classified as **malicious**.
  15. **Security Action:** The invention permits access to a legitimate destination, displays an explainable warning for a suspicious destination, and blocks or isolates a malicious destination.
  16. **Result Logging:** The invention records the source, final URL, redirect chain, detected indicators, risk score, classification result and security action.
  17. **Threat-Intelligence Feedback:** Confirmed malicious domains, URLs, redirect patterns and webpage indicators are added to the threat-intelligence repository.
  18. **Model and SIEM Update:** The stored intelligence supports model improvement, threshold adjustment and model version management. In an enterprise deployment, malicious-event information is transmitted to the SIEM system for monitoring and incident response.
7. **List out the features of your invention which are believed to be new and distinguish them over the closest technology.**

The following claims define the features of the present invention that are believed to be novel and distinguish it from the closest existing technologies.

**Claims:**

- **A QR-code phishing detection and prevention system**, comprising QR-code acquisition, image preprocessing, tampering detection, multimodal feature analysis, feature fusion, risk-score generation, intelligent classification and adaptive policy enforcement, wherein the QR code and its encoded destination are analysed before access to the destination is permitted.
- **The system as claimed in Claim 1**, wherein QR-image characteristics, encoded URL features, redirection behaviour, domain information, SSL/TLS certificate details and destination webpage content are jointly analysed to detect a QR-code phishing attack.
- **The system as claimed in Claim 1**, wherein QR-code overlays, physical replacement, structural irregularities, geometric distortion or digital manipulation are detected together with malicious destination behaviour.

- **The system as claimed in Claim 1**, wherein shortened URLs and intermediate redirections are traced in an isolated environment to identify the final destination without exposing the user's active browser session.
- **The system as claimed in Claim 1**, wherein the Multimodal Fusion Engine combines QR-image, URL, redirection, domain, certificate and webpage-content features to generate a dynamic Quishing Risk Score.
- **The system as claimed in Claim 1**, wherein a risk-classification threshold is dynamically adjusted according to the QR-code source, encoded-content type, application context, model confidence and available threat intelligence.
- **The system as claimed in Claim 1**, wherein the QR code is classified as legitimate, suspicious or malicious and the system respectively permits access, displays an explainable warning, or blocks and isolates the destination.
- **The system as claimed in Claim 1**, further comprising a persistent threat-intelligence feedback mechanism configured to record detection results and support model updates, threshold adjustment and SIEM-based security monitoring.

### **Principal Novelty of the Invention**

The principal novelty of the present invention lies in the unified and coordinated integration of pre-redirection QR-code interception, QR-image tampering detection, multimodal analysis of URL, redirection, domain, SSL/TLS certificate and webpage-content features, dynamic Quishing Risk Score generation, context-adaptive classification and automated security enforcement.

Unlike conventional approaches that generally analyse either the QR-code image or its encoded URL independently, the present invention jointly examines the QR code and its final destination before user access is permitted. The persistent threat-intelligence feedback mechanism further supports continuous improvement of the detection model, classification thresholds and organisational security monitoring.

8. **Has the invention been built or tested or implemented? If yes please provide the Efficiency/Efficacy details of the invention –(In Progress)**
9. **Briefly state when and how you first conceived this idea?**
10. **Have you sold, offered for sale, publicly used or published anything related to this invention? If yes, please briefly explain the dates and circumstances. List those individuals to whom you have revealed your invention. Were non-discloser documents signed before discloser in each case? Please state any deadlines of which you may be aware for filing an application on this invention.**
9. **Include any reasons that your invention would not have been obvious to someone of average skill in the art.**
10. **Additional comments by inventor (if you want to give more details out of scope of this IDF).**

The proposed invention provides preventive protection against QR-code phishing by analysing the QR code and its destination before permitting access. It may be implemented as a mobile application, browser extension, banking-security module, cloud API or enterprise security solution. The invention is not limited to a particular machine-learning algorithm and may support on-device, cloud or hybrid processing. Its threat-intelligence feedback mechanism enables continuous improvement using newly detected malicious URLs,

domains and QR-code patterns. The proposed architecture is a preferred implementation, and suitable modifications may be made without changing the essential scope of the invention.

## 11. Drawings/Flowchart/Table

